Computer Science & DA

Warehousing

Lecture - 05

By- Vishvadeep Gothi sir

Recap of Previous Lecture

Topic Data Transformation

Topic Data Cleaning

Topic Data Smoothing

Topics to be Covered

- **Topic** Data Smoothing
- **Topic** Exponential Smoothing
- **Topic** Data Normalization

Topic : Data Smoothing

1. Binning
2. Moving Average
3. Random walk
4. Random method
5. Exponential smoothing

Topic : Smoothing by Binning

50, 6, 2, 48, 42, 34, 29, 25, 15, 45, 15, 45, 49, 12, 25, 2

Make bins of 4 values

2, 2, 6, 12 | 15, 15, 25, 25 | 29, 34, 42, 45 | 45, 48, 49, 50

1. Smoothing by bin mean → 6, 6, 6, 6 20, 20, 20, 20 38, 38, 38, 38 48, 48, 48, 48
2. Smoothing by bin boundary

2, 2, 2, 12 15, 15, 25, 25 29, 29, 45, 45 45, 50, 50, 50

Topic : Smoothing by Moving Average

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

3 Period moving average

8	
16	
9	13
15	15
21	19
21	22
24	25
30	27
26	28
27	28
30	30
34	

$$\frac{8+16+9}{3} = 11$$

Topic : Smoothing by Moving Average

Date	Close
01-01-23	81.42
02-01-23	84.67
03-01-23	84.33
04-01-23	86.85
05-01-23	88.28
06-01-23	87.87
07-01-23	87.84
08-01-23	87.77
09-01-23	79.37
10-01-23	78.39
11-01-23	77.60
12-01-23	79.80
13-01-23	81.29
14-01-23	85.37
15-01-23	89.35
16-01-23	86.55

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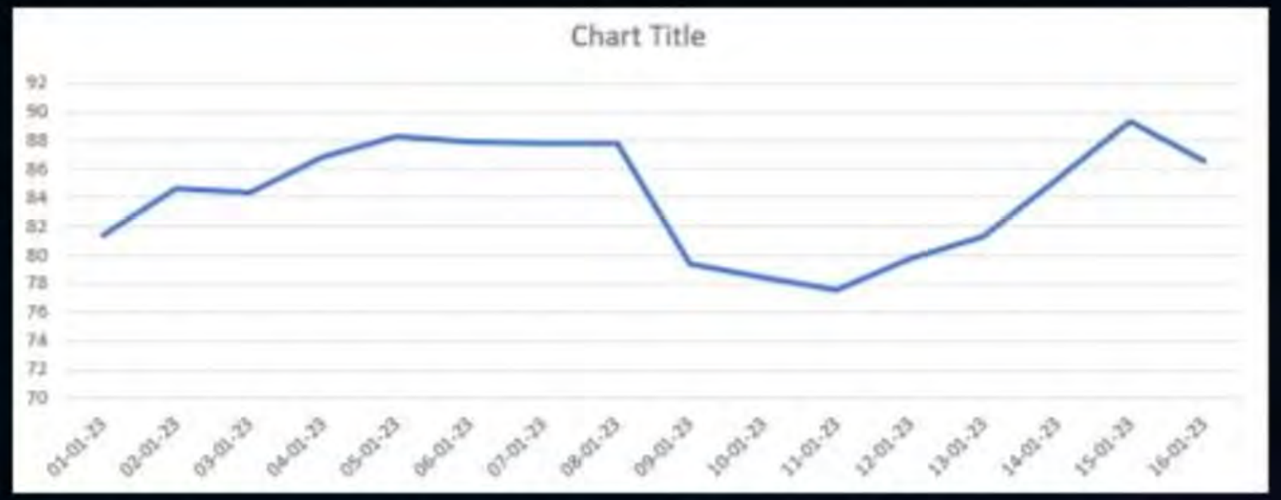

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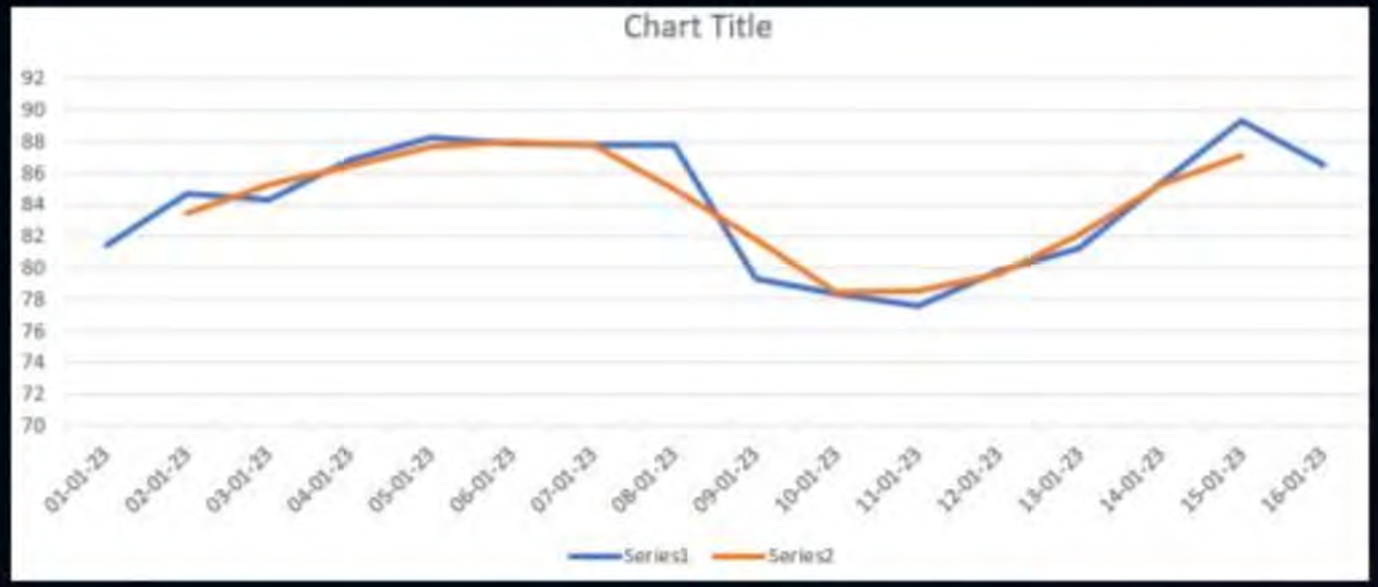
16-01-23	86.55	

Topic : Smoothing by Moving Average

Date	Close	3 Period Moving Average
01-01-23	81.42	
02-01-23	84.67	83.47333
03-01-23	84.33	85.28333
04-01-23	86.85	86.48667
05-01-23	88.28	87.66667
06-01-23	87.87	87.99667
07-01-23	87.84	87.82667
08-01-23	87.77	84.99333
09-01-23	79.37	81.84333
10-01-23	78.39	78.45333
11-01-23	77.60	78.59667
12-01-23	79.80	79.56333
13-01-23	81.29	82.15333
14-01-23	85.37	85.33667
15-01-23	89.35	87.09
16-01-23	86.55	



Topic : Smoothing by Moving Average

Date	Close	3 Period Moving Average	7 Period Moving Average
01-01-23	81.42		
02-01-23	84.67	83.47333	
03-01-23	84.33	85.28333	
04-01-23	86.85	86.48667	
05-01-23	88.28	87.66667	
06-01-23	87.87	87.99667	
07-01-23	87.84	87.82667	
08-01-23	87.77	84.99333	
09-01-23	79.37	81.84333	
10-01-23	78.39	78.45333	
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Topic : Smoothing by Moving Average

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03-01-23	84.33	85.28333	
04-01-23	86.85	86.48667	85.89429
05-01-23	88.28	87.66667	86.80143
06-01-23	87.87	87.99667	86.04429
07-01-23	87.84	87.82667	85.19571
08-01-23	87.77	84.99333	83.87429
09-01-23	79.37	81.84333	82.66286
10-01-23	78.39	78.45333	81.72286
11-01-23	77.60	78.59667	81.37
12-01-23	79.80	79.56333	81.59571
13-01-23	81.29	82.15333	82.62143
14-01-23	85.37	85.33667	
15-01-23	89.35	87.09	
16-01-23	86.55		

Topic : Smoothing by Moving Average

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09-01-23	79.37	81.84333	82.66286
10-01-23	78.39	78.45333	81.72286
11-01-23	77.60	78.59667	81.37
12-01-23	79.80	79.56333	81.59571
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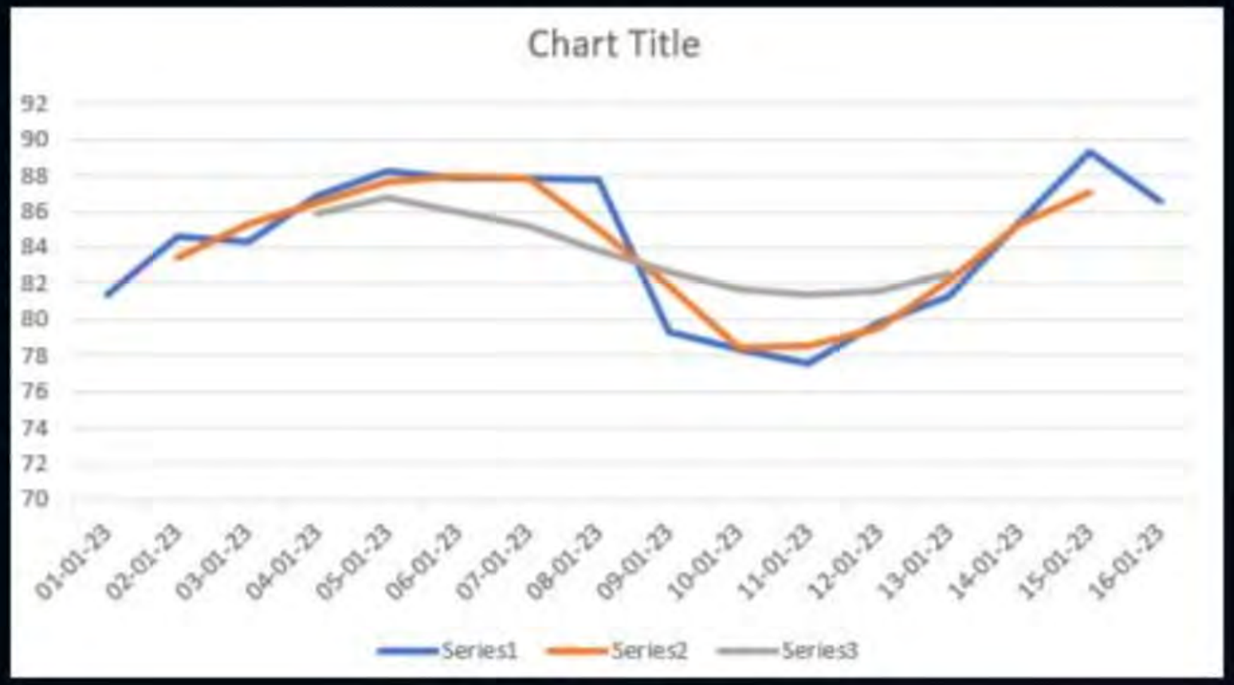


Topic : Smoothing by Moving Average

Date	Close	4 Period Moving Average	Centering
01-01-23	8		
02-01-23	21		
03-01-23	13	15	$\frac{15+15}{2} = 15$
04-01-23	17	15	$\frac{15+16}{2} = 16$
05-01-23	9	16	$\frac{16+18}{2} = 17$
06-01-23	25	18	$\frac{18+20}{2} = 19$
07-01-23	20	20	$\frac{20+23}{2} = 22$
08-01-23	27	23	
09-01-23	19		

Topic : Smoothing by Moving Median

Date	Close	3 Period Moving Median
01-01-23	8	
02-01-23	21	13
03-01-23	13	17
04-01-23	17	13
05-01-23	9	17
06-01-23	25	20
07-01-23	20	25
08-01-23	27	20
09-01-23	19	

Topic : Smoothing by Moving Average

Date	Close	4 Period Moving Median	Centering
01-01-23	8		
02-01-23	21		
03-01-23	13	15	
04-01-23	17	15	15
05-01-23	9	15	15
06-01-23	25	19	17
07-01-23	20	23	21
08-01-23	27	23	23
09-01-23	19		

8, 13, 17, 21

$$\text{median} = \frac{13+17}{2} = 15$$

Topic : Exponential Smoothing

If the data has no trend and no seasonal pattern, then this method is essentially used

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

This method uses weighted moving averages with exponentially decreasing weights

Sales

Actual (A)	Forecast (F)
1.89 cr	2 cr. ← (t-1)
	◯ ← t

F

$$\alpha * A_{t-1} + (1-\alpha) F_{t-1}$$

$$\alpha A_{t-1} + F_{t-1} - \alpha F_{t-1}$$

$$F_{t-1} + \alpha (A_{t-1} - F_{t-1})$$

values V_1, V_2

$$\text{avg.} = \frac{V_1 + V_2}{2}$$

$$= \frac{V_1}{2} + \frac{V_2}{2}$$

$$= 0.5V_1 + 0.5V_2$$

$$\Downarrow$$

weighted avg.

$$= 0.6V_1 + 0.4V_2$$

or

$$= 0.7V_1 + 0.3V_2$$

or

$$= 0.2V_1 + 0.8V_2$$

3

values V_1, V_2, V_3

$$\text{avg} = \frac{V_1 + V_2 + V_3}{3}$$

$$= 0.33V_1 + 0.33V_2 + 0.33V_3$$

Topic : Exponential Smoothing

Week	Actual Sales A	Smooth Value (Forecast) α=0.2 F	Smooth Value (Forecast) α=0.4
1	39	39	
2	44	0.2*39 + 0.8*39 = 39	
3	40	= 0.2*44 + 0.8*39 = 40	
4	45	= 40	
5	38	= 0.2*45 + 0.8*40 = 41	
6	43	= 0.2*38 + 0.8*41 = 40	
7	39	= 0.2*43 + 0.8*40 = 41	
8		= 0.2*39 + 0.8*41 = 41	

Topic : Exponential Smoothing

α is low: give more weight to historical data

α is high: give more weight to Latest data.

Topic : Exponential Smoothing

Homework

Week	Actual Sales	Smooth Value (Forecast) α=0.4
1	310	
2	365	
3	395	
4	415	
5	450	
6	465	
7	475	
8		

Topic : Data Normalization

Data normalization involves converting all data variables into a given range

Normalization Techniques:

- Min-Max Normalization
- Z-score Normalization
- Decimal Scaling

ex :- Employee

Eid	Ename	age	Salary
E1	n1	25	2500000
E2	n2	26	2000000

f(age, salary)

Topic : Data Normalization

Why Data Normalization

PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
892	3	Kelly, Mr. J	male	34.5	0	0	330911	7.8292		Q
893	3	Wilkes, Mr	female	47	1	0	363272	7		S
894	2	Myles, Mr.	male	62	0	0	240276	9.6875		Q
895	3	Wirz, Mr. A	male	27	0	0	315154	8.6625		S
896	3	Hirvonen,	female	22	1	1	3101298	12.2875		S
897	3	Svensson,	male	14	0	0	7538	9.225		S
898	3	Connolly, I	female	30	0	0	330972	7.6292		Q
899	2	Caldwell,	male	26	1	1	248738	29		S
900	3	Abrahim,	female	18	0	0	2657	7.2292		C
901	3	Davies, Mr	male	21	2	0	A/4 48871	24.15		S
902	3	Ilieff, Mr.	male		0	0	349220	7.8958		S
903	1	Jones, Mr.	male	46	0	0	694	26		S
904	1	Snyder, Mr	female	23	1	0	21228	82.2667	B45	S
905	2	Howard,	male	63	1	0	24065	26		S
906	1	Chaffee,	female	47	1	0	W.E.P. 57	61.175	E31	S
907	2	del Carlo, I	female	24	1	0	SC/PARIS	27.7208		C
908	2	Keane, Mr	male	35	0	0	233734	12.35		Q
909	3	Assaf, Mr.	male	21	0	0	2692	7.225		C
910	3	Ilmakanga	female	27	1	0	STON/O2.	7.925		S
911	3	Assaf Khal	female	45	0	0	2696	7.225		C
912	1	Rothschild	male	55	1	0	PC 17603	59.4		C
913	3	Olsen, Ma	male	9	0	1	C 17368	3.1708		S
914	1	Flegenhei	female		0	0	PC 17598	31.6833		S
915	1	Williams,	male	21	0	1	PC 17597	61.3792		C
916	1	Ryerson,	female	48	1	3	PC 17608	262.375	B57 B59 B	C
917	3	Robins, Mr	male	50	1	0	A/5. 3337	14.5		S

Topic : Data Normalization

Min-Max Normalization

It is 0-1 normalization technique

normalizes all values between 0 to 1

Topic : Data Normalization

Min-Max Normalization

$$Value' = \frac{Value_i - min}{Max - Min}$$

min = 12
max = 23

max − min = 23 − 12 = 11

Values	Normalized Value
12	$\frac{12-12}{11} = 0$
14	$\frac{14-12}{11} = 0.2$
16	$\frac{16-12}{11} = 0.4$
18	$\frac{18-12}{11} = 0.5$
19	$\frac{19-12}{11} = 0.6$
23	$\frac{23-12}{11} = 1$

Topic : Data Normalization

Min-Max Normalization

$$Value' = \frac{Value_i - min}{Max - Min}$$

Values	Normalized Value
100	0
200	0.625
150	0.3125
170	0.4375
260	1
230	0.8125
160	0.375

Topic : Data Normalization

Standard Deviation

A quantity expressing by how much the members of a group differ from the mean value for the group

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

Topic : Data Normalization

$$\mu = \frac{12+14+16+18+19+23}{6} = 17$$

Standard Deviation

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

Values	$(x_i - \mu)$	$(x_i - \mu)^2$
12	-5	25
14	-3	9
16	-1	1
18	1	1
19	2	4
23	6	36

$$\sigma = \sqrt{\frac{25+9+1+1+4+36}{6}}$$

$$= 3.6$$

Topic : Data Normalization

Z-score Normalization

A statistical measurement that describes a value's relationship to the mean of a group of values

Topic : Data Normalization

Z-score Normalization

$$z_i = \frac{x_i - \mu}{\sigma}$$

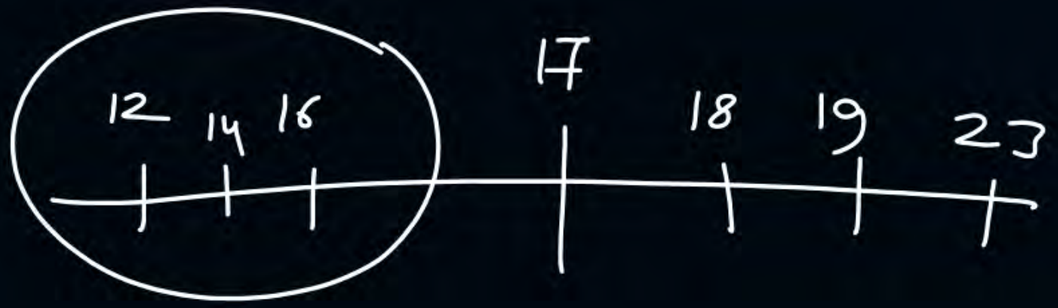
Topic : Data Normalization

Z-score Normalization

$$z_i = \frac{x_i - \mu}{\sigma}$$

$\mu = 17$

$\sigma = 3.6$



Values	$(x_i - \mu)$	[redacted]	Z
12	-5	[redacted]	-1.39
14	-3	[redacted]	-0.83
16	-1	[redacted]	-0.28
18	1	[redacted]	0.28
19	2	[redacted]	0.56
23	6	[redacted]	1.67

Homework Z – Score

Values	
10	
26	
23	
4	
15	
19	
27	
30	
21	
25	

2 mins Summary

Topic Data Smoothing

Topic Exponential Smoothing

Topic Data Normalization

Happy Learning

THANK - YOU